

Abstract

This report analyzes three supervised classifiers—RBF Support Vector Machine (SVM), Decision Tree, and k-Nearest Neighbors (KNN)—on four binary classification datasets stored in MAT format. The notebook first inspects the datasets, visualizes their two-dimensional structure with scatter plots, and then evaluates the classifiers using a manually implemented 5-times 2-fold cross-validation protocol. Performance is summarized through mean accuracy, average classifier ranking, and a Friedman statistical test. The reported results show that RBF SVM achieves the best overall mean accuracy (0.9467), followed by KNN (0.9343), while Decision Tree obtains the lowest average performance (0.9129). The Friedman test statistic of 7.6 with $p = 0.0224$ indicates a statistically significant difference among the three models at the 5% significance level.

1. Introduction

The work addresses a classical model-comparison task in supervised machine learning. The goal is to study how different learning algorithms behave on small two-dimensional datasets with varying class balance and class separation. By using multiple datasets and repeating the train/test partition five times, the analysis aims to reduce the risk of drawing conclusions from a single favorable split. The workflow follows a complete experimental pipeline: data loading, exploratory inspection, model definition, repeated cross-validation, statistical testing, and graphical presentation of the results.

2. Workflow and methodology

I imported NumPy, Pandas, SciPy, Matplotlib, and Scikit-learn utilities. Four datasets are loaded from Google Drive using `scipy.io.loadmat`. Each dataset contains two input features and binary labels. The analysis begins with a shape and class preview for the first dataset and then constructs a compact overview table for all four datasets. After that, the notebook visualizes every dataset in a 2×2 scatter plot grid, which helps interpret classifier performance in terms of geometric class structure.

2.1. Models used

Three classifiers are evaluated. The first is an RBF SVM implemented as a pipeline with `StandardScaler()` followed by `SVC(kernel="rbf", C=1.0, gamma="scale")`. The second is a Decision Tree with Gini impurity and maximum depth 20. The third is a KNN classifier, also preceded by standardization, with 3 neighbors and Euclidean distance. Scaling is correctly applied to SVM and KNN because these methods are distance- or margin-based, whereas the tree is used without scaling.

2.2. Evaluation protocol

For each dataset, the notebook performs a manual 5×2 cross-validation scheme. At every repetition, indices are randomly permuted within each class so that the split remains stratified. Half of the samples of each class are assigned to the training fold and the remaining half to the test fold. Each classifier is trained twice per repetition: first on split A and evaluated on split B, then on split B and evaluated on split A. The two accuracies are averaged, and the final dataset score is the mean of the five repeated averages. This procedure provides a more stable estimate than a single train/test split.

Dataset overview

The four datasets differ in size and class distribution. Datasets 1 and 2 each contain 300 samples with a 1:2 class ratio. Dataset 3 contains 600 samples with the same 1:2 ratio, while Dataset 4 contains 600 samples and is class-balanced. Because all datasets have exactly two features, the visual scatter plots are directly interpretable.

Dataset	Samples	Features	Class 1	Class 2	Class 2 ratio
dataset1	300	2	100	200	0.6667
dataset2	300	2	100	200	0.6667
dataset3	600	2	200	400	0.6667
dataset4	600	2	300	300	0.5000

Table 1: Summary of the four datasets used in the notebook.

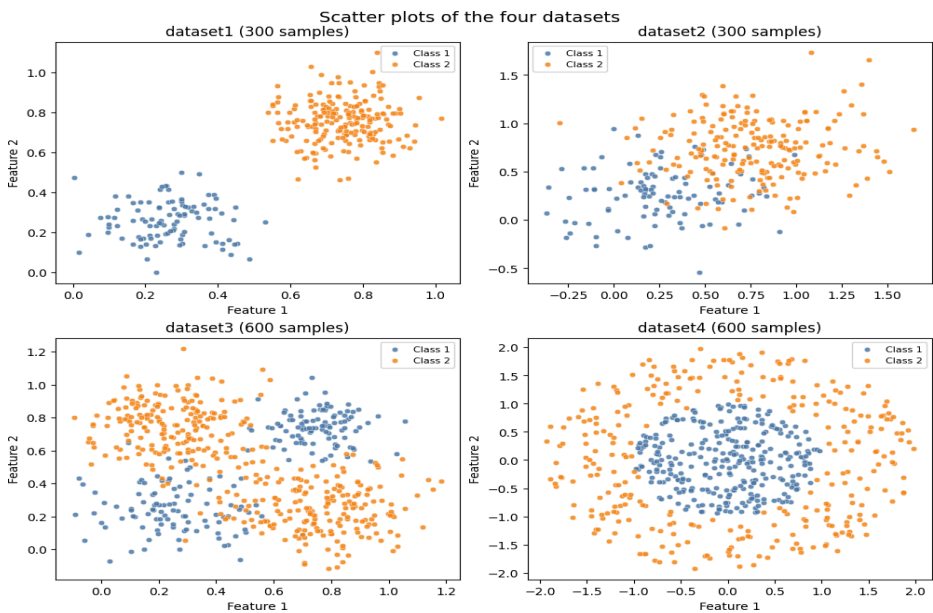


Figure 1: Scatter plots of the four binary datasets. The visual structure suggests that dataset2 is the most challenging, while datasets1 and 4 appear easier to separate.

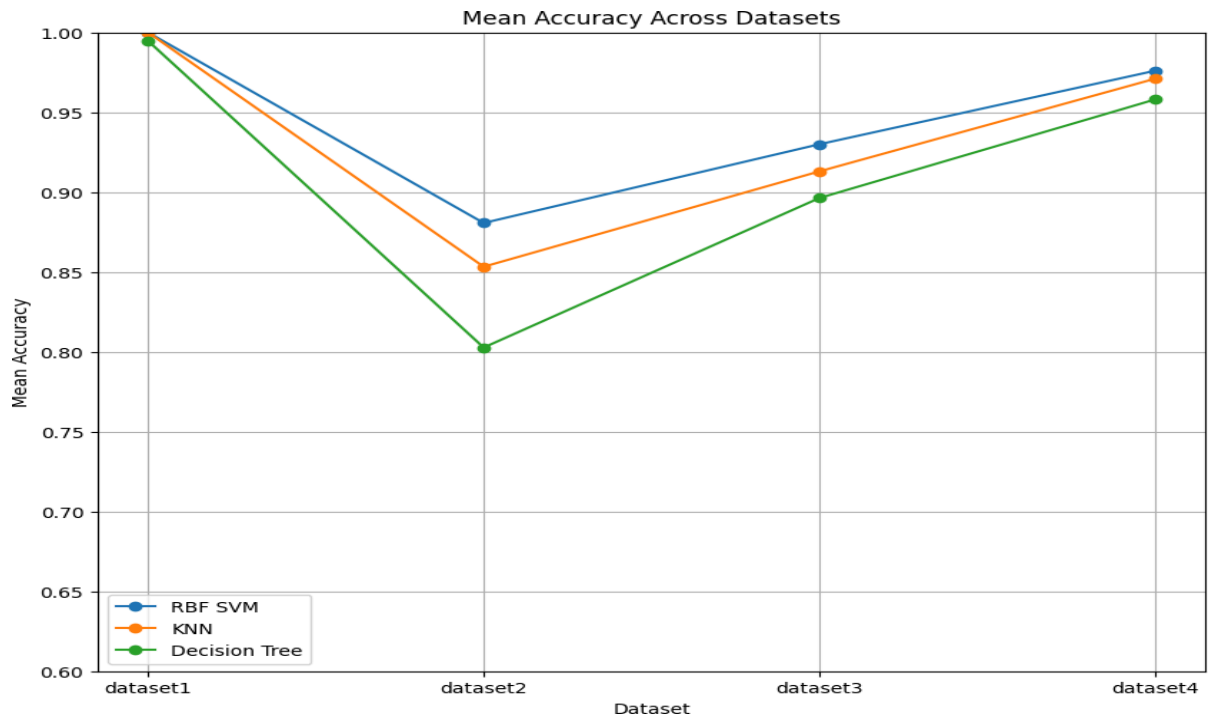


Figure 2: Mean accuracy trend across datasets for the three classifiers.

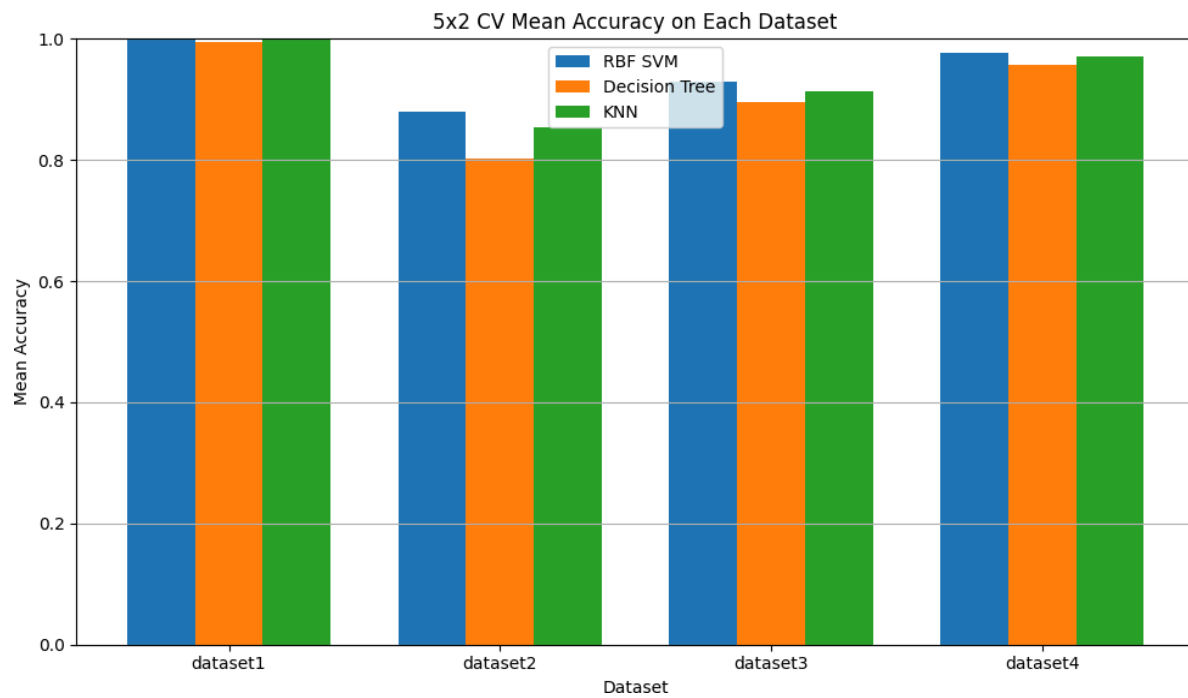


Figure 3: Bar chart comparing 5×2 cross-validation mean accuracy on each dataset.

Quantitative results

This result stores one mean 5×2 cross-validation accuracy per classifier and per dataset. The resulting matrix shows consistently strong performance for RBF SVM and KNN, with Decision Tree trailing slightly in every case. Dataset2 is the most difficult benchmark for all methods, whereas datasets1 and 4 are solved with very high accuracy. The full average accuracy table is reported below.

Dataset	RBF SVM	Decision Tree	KNN
dataset1	1.0000	0.9947	1.0000
dataset2	0.8807	0.8027	0.8533
dataset3	0.9300	0.8963	0.9130
dataset4	0.9760	0.9580	0.9710
Mean	0.9467	0.9129	0.9343

Table 2: Mean accuracy obtained by each classifier on each dataset and overall mean across datasets.

1.1 Statistical comparison and ranking

To verify whether the observed differences are likely due to random variation, the notebook applies the Friedman test to the per-dataset mean accuracies. The test returns a statistic of 7.6 and a p-value of 0.0224. At $\alpha = 0.05$, the null hypothesis of equal classifier performance is rejected. The average ranks confirm this ordering: RBF SVM = 1.125, KNN = 1.875, and Decision Tree = 3.000. Therefore, RBF SVM is the most competitive method in this experiment, with KNN as the second-best alternative.

Interpretation

The results are consistent with the geometry suggested by the scatter plots. The RBF SVM performs best because it can model non-linear decision boundaries while still benefiting from feature scaling. KNN also adapts well to local structure, but it is slightly more sensitive to class overlap and neighborhood ambiguity. The Decision Tree is competitive on easier datasets but appears less stable on the harder ones, especially dataset2, where it records the lowest score. The balanced class distribution of dataset4 may also contribute to the strong and stable performance observed for all methods.

Limitations and possible improvements

Although the notebook is well structured and the comparison is meaningful, some limitations remain. First, only accuracy is reported, so the analysis does not discuss precision, recall, or confusion matrices. Second, the Friedman test identifies a significant overall difference but does not include a post-hoc test to determine which pairs of classifiers differ significantly. Third, the hyperparameters are fixed rather than optimized, so the comparison evaluates default design choices instead of best-tuned models. Future work could add repeated stratified cross-validation using Scikit-learn utilities, hyperparameter search, and post-hoc statistical analysis such as Nemenyi or Wilcoxon signed-rank comparisons.